

Course: Transversal ISI Project (ING4187)

Project 2: Kingdom Communication Solutions (KCS) Media Studio - Web-Based AI Video Transcription and Multilingual Caption Editing

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1 Chapter 1: Introduction

1.1 General Background and Trends

In the contemporary digital era, multimedia communication has become a predominant vehicle for global knowledge dissemination and education [1]. Advances in artificial intelligence, large language models, and self-supervised neural speech representations have catalyzed automated video localization pipelines capable of transcribing, subtitling, and translating audiovisual content at scale [2, 3]. Modern architectures increasingly embed multimodal workflows that automatically align phonemes and perform cross-lingual representation learning across widespread languages [4–6]. However, these automated workflows overwhelmingly serve commercially dominant languages, relying on massive supervised corpora that remain scarce for the vast majority of indigenous African tongues. In multilingual sub-Saharan regions, the absence of tailored digital authoring tools disenfranchises non-Western linguistic communities, reinforcing an acute communicative divide where current technologies fail to support tonal variations, complex orthographies, and regional dialects [7–9].

1.2 Motivation

The Republic of Cameroon represents a highly dense linguistic ecosystem comprising over two hundred and fifty distinct mother tongues, including Ghɔmálá', Ewondo, and Basaa. Despite this cultural richness, local populations face systemic exclusion from digital multimedia due to the lack of computational resources and standardized digital corpora for these languages [10]. Two primary motivations guide this research. **MS1:** There is a critical need for reliable, validated speech and text resources for low-resource Cameroonian languages, because the lack of standardized corpora severely limits the development of effective automatic speech recognition and multimedia localization technologies, thereby threatening the preservation of the country's intangible linguistic heritage [11, 12]. **MS2:** Enhancing socioeconomic and civic inclusivity requires assistive video editing technologies, yet existing web-based transcription and caption-editing environments are not sufficiently adapted to the regional linguistic variation, diacritics, and specific orthographic requirements of low-resource Cameroonian languages.

1.3 Overall Goal and Problem Statements

The overall goal of this research is to design, develop, and evaluate a web-based AI-assisted video transcription and multilingual caption-editing platform, known as KCS Media Studio, that natively supports Cameroonian languages while integrating validated linguistic resources and adaptive mechanisms for handling regional language variation. Achieving this goal requires overcoming significant technical hurdles highlighted by two core problem statements. **PS1:** Existing speech recognition and transcription systems are fundamentally constrained by the limited availability of standardized, validated speech datasets, phonological dictionaries, and acoustic models for tonal Cameroonian mother tongues, rendering mainstream algorithms prone to high error rates. **PS2:** Conventional video transcription and caption-editing environments do not adequately accommodate regional accents, slang, dialectal variation, and timeline desynchronization issues inherent to the General Alphabet of Cameroonian Languages, forcing media creators into labor-intensive manual timeline adjustments that fail to reflect the cultural nuance of native speech [13].

1.4 Research Questions

To systematically address these sociolinguistic and technical barriers, this study investigates two central research questions. **RQ1:** How can the editing environment accommodate variations in regional accents or slang without alienating users from different villages or communities who speak the same base language? **RQ2:** How can a multilingual speech-transcription tool be developed so that users can smoothly edit videos while integrating written lexical resources and voice recordings for low-resource Cameroonian languages seamlessly within a browser application?

1.5 Methodology and Research Objectives

This research formally adopts the multi-methodological information systems development framework established by Nunamaker et al. [14], which bridges theoretical modeling and empirical observation through four phases: Observation, Theory Building, Systems Development, and Evaluation. Rather than merely following a generalized software lifecycle, this framework is intimately linked to the KCS Media Studio project by ensuring that every conceptual algorithm for handling tonal variations directly translates into an engineered platform feature, which is then strictly evaluated by native speakers. Following this framework, the research questions are decomposed into specific Research Objectives (**ROs**). For **RQ1**, the objectives are to investigate existing transcription systems regarding dialectal and orthographic handling (**O1.1.O**); develop a conceptual design for an adaptive editing environment incorporating localized Cameroonian vocabulary (**O1.2.T**); implement these adaptive caption functionalities within the KCS Media Studio prototype (**O1.3.I**); and evaluate whether the environment effectively supports regional variation for the intended indigenous users (**O1.4.E**). For **RQ2**, the objectives are to examine multilingual speech approaches and requirements for integrating lexical data into web workflows (**O2.1.O**); design an architecture connecting audio recordings, lexicons, and video caption editing (**O2.2.T**); implement the proposed transcription and resource integration components (**O2.3.I**); and finally assess transcription performance, integration reliability, and overall user experience through controlled trials (**O2.4.E**).

1.6 Approach and Outline of the Work

The methodological approach begins with community-based corpus acquisition using structured mobile forms to compile aligned audio pairs and core vocabulary across the target pilot languages [15]. These corpora directly inform the fine-tuning of self-supervised multilingual representations. The software artifact is then engineered as a cloud-native platform featuring an automated rendering backend and an intuitive frontend, which is ultimately validated through controlled laboratory trials with native translators measuring throughput against traditional baselines [16]. The remainder of this work is structured as follows. Chapter 2 presents the State of the Art regarding low-resource speech technologies and digital linguistic resources. Chapter 3 outlines the system design and architecture of the platform. Chapter 4 details the proof-of-concept implementation. Chapter 5 presents the evaluation methodology alongside the experimental results. Finally, Chapter 6 concludes the study by summarizing the main contributions and outlining potential avenues for future work.

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